

Rabin Cryptosystem Based Biometric Privacy-Preserving User Authentication Scheme for IoT Devices over Cloud

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Abstract—The uses of smart devices are growing rapidly in recent times due to the wide range of IoT based applications. Security is one of the major concerns for IoT devices and its applications. The device security can only be warranted through robust multi-factor device authentication mechanisms. Failure in authentication, might lead the system to feed the incorrect data and may lead to dangerous attacks. As a result, IoT enable devices require secure authentication schemes. In this paper, an user authentication scheme known as Rabin Cryptosystem based Biometric Privacy-Preserving User Authentication Scheme (RCBP2U-AS) is proposed. The proposed scheme, compared to other schemes, is quite better at providing security and authentication. Using the Automated Validation of Internet Security Protocols and Applications (AVISPA) simulation tool, it is found that the system is resistant to replay, man-in-the-middle attacks, impersonation attacks, password guessing attacks and many more. The proposed algorithm is evaluated in terms of computation, communication, storage overhead, and overall computation time.

Index Terms—cloud server, authentication, security, AVISPA, Rabin cryptosystem, internet of things, cloud computing, biometrics

I. INTRODUCTION

Internet of Things (IoT) is an emerging technology in which the applications are based on context-aware and ubiquitous in nature [4]. It enables access to multiple heterogeneous devices in which the devices are distributed spatially and its networks provides numerous services to the user. This is considered to be one of the futuristic things in LAN, RFID for its intelligibility, greater recognition, control-ability towards various networks [13]. IoT applications have varied usages in healthcare, security and surveillance, traffic management systems and other areas. These systems collect data from various sources through actuators, sensors and other smart devices. The networks which are linked to the IoT devices yields exceptional volume of data [10]. Typically the information are stored in cloud [12]. There are three kinds of cloud services: (i) Infrastructure as a service (IaaS), (ii) Platform as a Service (PaaS), and (iii) Software as a service (SaaS). In

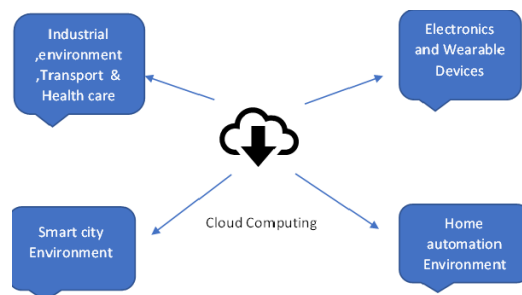


Fig. 1. Authentication deployment needed over Cloud based IoT system

IaaS, which is also considered as the basic layer in the cloud, provides various service to the end users on demand. Second service (PaaS), which yields control to access the information related to IoT. In order to monitor the application domains, Software as a service (SaaS) is used. Moreover, the end user gets unlimited access to the application they are using [19]. Using the cloud storage user can able to access the information anytime and from everywhere. However, the data are secured by authorized controls and only owner of the data can view the information stored. In cloud storage security is the major concern as attackers may target the storage information by breaking the access.

A cloud based IoT system that required authentication deployment is shown in Figure 1. As shown in this figure, different IoT environments are interconnected with cloud server. Any user wants to access the data from cloud server needs to pass through cloud authentication mechanism. In applications like smart healthcare, the remote user acts as a health expert, here the patient and the expert gets connected through the IoT while the patient is accessing these services remotely from home. But in some cases, it causes various security concerns related to confidentiality and authenticity [14]. The major concern about the legitimacy is that, the end user should

be validated with their own information, such as the user's type, information they knows and etc. Here, the type of users comes under their physical attributes such as finger prints and related bio-metric stuffs, and the information which the user belongs, such as smart card, and the information known is their personal password [7]. The scheme of authenticity verifies the information with one another in the sense, the server and the remotely active user produces a session which can be secured to transfer data [1]. In this paper an authentication scheme known as Rabin cryptosystem based biometric Privacy-Preserving User Authentication Scheme (RCBP2U-AS) is proposed. It consisting of (i) key generation, (ii) encryption, and (iii) decryption. The technique comprises total of six stages: Pre-deployment, registration, user-Login, password changes, and dynamic addition of nodes. The proposed technique is more secure to prevent from different attacks such as: man in the middle attack, password guessing and impersonation attacks, replay and some others. The rest part of the paper is structured as follows. Related work is discussed in Section II, and proposed scheme in Section III. Result analysis and conclusions in Section IV and V respectively.

II. RELATED WORK

There has been significant work carried in the direction of privacy preserving security solutions. As per most recent surveys, implementation of mutual authentication between a user and device has been substantial towards identifying the challenges during handshake process. Few related work on IoT security are discussed here. An authentication scheme for wireless sensor networks without the support of the gateway nodes is presented. In this work an authenticated user can access the sensor data. It uses XOR function for its implementation and the scheme is considered to be very delicate [15]. An untraceability issues with few attacks like stolen card attacks, man-in-the middle attacks are addressed and performance analysis demonstrated using proVerif [7]. As highlighted on advanced key agreement technique for authentication using lightweight solution over IoT platform. The method uses heuristic approach for security analysis for demonstrating security properties that are secure against all possible attack under internet integrated WSN environment [8]. A lightweight scheme proposed by [2] the technique lacks in providing security against attacks such as (i) off-line password guess attacks, (ii) impersonation, and (iii) sensor node impersonation attacks in addition, still issues are in existence with Energy efficiency. In WSN platform sensor node lifetime is a major concern as it can affects the lifetime of networks. To enhance the sensors lifetime and its data [18] a scheme is proposed to improvised authentication technique [17]. The solution against attacks such as session-specified temp attack, brute force and smart card attacks with improvised authentication scheme using RSA is proposed and found to be appropriate. [9]. A new scheme presented on authentication for sensor network includes random oracle model and protocol analysis that provides security against sensor node physical capturing, insider, user's forgery and offline guessing attacks [17]. As

Elliptic Curve Cryptography (ECC) is considered as more optimal than the RSA and RSA doesn't have longest key length [20] [3]. Due to this reason at WSN platforms several ECC based authenticated system have been identified that provides good performance [16]. The scheme addresses synchronization related issues and information attacks. The strong password protections technique with biometric authentication system requirement remains challenging to accomplish on various IoT platforms [6]. Cloud security for IoT enabled devices, includes authentication scheme which is improved in features established. The authors emphasised that proposed system lacks in identity hidden attributes and offline wrong guessing attacks. Also, this scheme could not able to fulfil the audit feature and wrong guessed attacks [3]. Moreover, along with the IoT activated devices over cloud computing platform a new method suggested that implements a threshold concept for distribution of session key and hash based password based confidential communication platform [11]. A new lightweight system-based on cloud proposed by an author scheme known as LAM-C IoT that uses fuzzy extractor to perform biometric verification for user authentication with device. The AVISPA tool employed to perform security analysis using Real-OR-Random (ROR) Model [16]. IoT is considered to be cloud computing next persuasive innovation everywhere. Its applications are wide spread in healthcare and other industries as well to remotely access the IoT data through cloud. Although it has significant advantages all over the industry it faces threats like access control, identity management threats and security threats as well. Hence this paper includes contribution in developing authentication scheme well equipped with security features for IoT enabled devices.

III. PROPOSED METHODOLOGY

The authentication techniques are not that enriched in order to utilize in the internet of things platform. However, these techniques are not sufficient to perform the high-level security in cloud. Since, the client and the server performed in unknown platform, thus privacy seems to be crucial at this point. Here, the information has been managed by the cloud server that can be accessed by anyone. Thus, in case if full authenticity is required to unlock the access, the authentication of the cloud server should be implemented in such a manner so that it should prevent from all possible attack and stealing of information. Likewise, the networking security should also be enhanced in a trust-able environment.

For this, the major contribution in the paper, a technique Rabin cryptosystem based biometric Privacy-Preserving User Authentication Scheme (RCBP2U-AS) has been implemented for cloud.

The technique includes three processing elements to generate the key, encryption decryption. Also some of the tasks are needed to be performed. Between the gateway node (GW) and smart device SD_j , a paired key is implemented. Now to form the secret key duration between U_i & SD_j a smart device (SD_j) for mutual validation the external party is created securely for the upcoming communication. The following

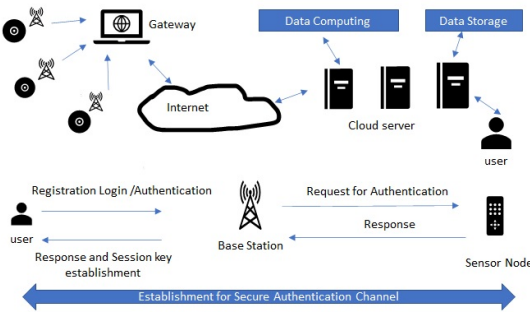


Fig. 2. RCBP2U-AS Work flow Model

TABLE I
TERMINOLOGY USED IN RCBE-AS

Symbol	Description
U_i	User
GW	Gateway node
S_j	Sensor node
ID_i	Identity of user
Bio_i, ID_i, SC_i	U_i smart card, identity and biometrics
ΔT	Password of user
$Gen(), Rep()$	Fuzzy extraction probabilistic generation & deterministic reproduction process
A	Adversary
bk_i, pr_i	Public reproduction parameter & U_i biometric secret key
$HamDit(BIO'_i, BIO_i)$	Hamming distance between two biometrics (BIO'_i, BIO_i)
SD_j	Identity of sensor node
RTS_i, RID_{SD_j}	Registration timestamps of U_i and SD_j at the GW
T_1, T_2, T_3	Current timestamps
TC_{gw}, TC_i	Temporal credentials of U_i and SD_j at the GW
$h(.)$	One-way hash function

steps contains the user validation part. Firstly, registration with one of the offline smart devices. Secondly, registration with user. Third, Login authentication and then the addition of the fresh smart device, after that the update of the biometric. Also, the Gateway (GW) and smart devices are then established with key pairs during the authentication part.

A. Offline Smart Device Registration Phase

The step usually occurs whenever the IoT devices are implemented in the cloud environment. In this case, the test performed on offline with the GW trusted medium to load the needed information at earlier into the storage of every smart device before they deploy. As shown in Figure 3. The algorithm maintains the strict constraint to proceed next level of verification with both end device and user. All the contents are sent in encrypted manner.

To complete this phase, one must go through the following steps of:

Step 1: The GW chooses a unique identity ID_{SD_j} and a temporary identity TID_{SD_j} for each deployed smart device SD_j , as well as a unique secret key_{SD_j} . The GW also creates

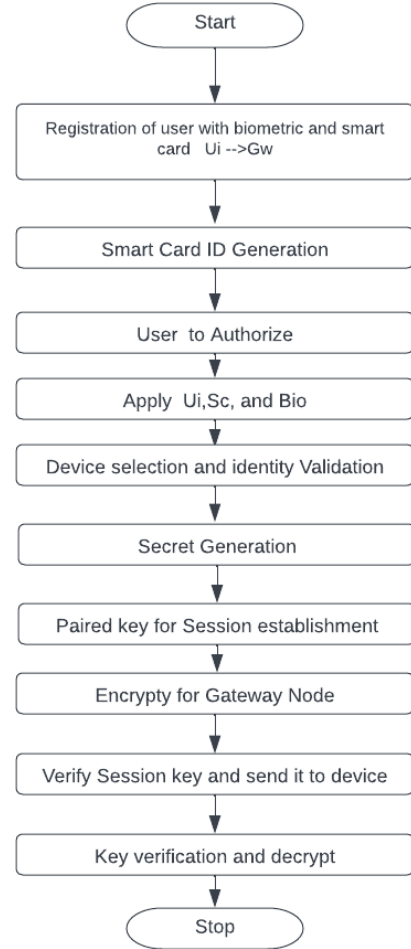


Fig. 3. Flowchart of proposed Authentication Scheme

its own unique identity ID_{gw} , temporary identity TID_{gw} , secret key d_{gw} . also, a master key k .

Step 2: For every SD_j , the GW further computes the pseudo-identity $RID_{SD_j} = h(ID_{SD_j} || k)$ and the temporal credential $TC_{SD_j} = h(RID_{SD_j} || ID_{gw} || d_{gw} || RTS_{SD_j})$, where $h(.)$ is a collision-resistant cryptographic hash function and RTS_{SD_j} is SD_j registration timestamp.

Step 3: The GW produces a t - bivariate symmetric polynomial of degree $f(m, n) = \sum_{i,j=0}^t a_{ij} m^i n^j \in GF(p)[m, n]$, where a_{ij} are coefficients chosen from a finite field $GF(p) = Z_p, Z_p = \{0, 1, \dots, p-1\}$ and p is a huge number, say 160 bits. We have $f(m, n) = f(n, m)$ because $f(m, n)$ is symmetric. The GW then calculates the polynomial share SD_j as $f(TID_{SD_j}, n) = \sum_{i,j=0}^t a_{ij} (TID_{SD_j}^i) n^j$, it is a univariate polynomial of t - degrees whose coefficients originate from again $GF(p)$. Furthermore, the GW also calculates its own polynomial share $f(TID_{gw}, n) = \sum_{i,j=0}^t a_{ij} (TID_{gw}^i) n^j$.

Step 4: lastly, every SD_j is preloaded with the data $\{dSD_j, TCSD_j, (TID_{SD_j}, RTS_{SD_j}), f(TID_{SD_j}, n)\}$. Before it is deployed in the deployment phase. The GW also saves its own data $\{k, Id_{gw}, d_{gw}, TID_{gw}, f(TID_{gw}, n)\}$ as well as data $\{(TID_{SD_j}, RTS_{SD_j})\}$ corresponding to each SD_j .

B. User Registration Phase

The GW handles registration on user by utilizing a safer medium for instance a person. The registration of a user U_i at the GW needs below steps.

Step 1: U_i selects an identity ID_i of his or her choosing, and then imprints personal biometric information, such as BIO_i on a terminal's sensor. In some papers, fuzzy extractor approach has been utilized for biometric authentication.

Step 2: $U_i \Rightarrow GW \leq RID_i > U_i$ then selects a secret key d_i and calculates its pseudo-identity $RID_i = h(ID_i || d_i)$, before securely sending the registration request $\{RID_i\}$ to the GW.

Step 3: $GW \Rightarrow U_i \leq SC_i >$ when the GW receives a registration request from U_i , it computes the temporal credential for U_i as $TC_i = H(RID_i || d_{gw} || RTS_i)$, where RTS_i , U_i the registration timestamp. The GW chooses TID_i as the U_i temporary identity. Furthermore, the GW generates its own temporal credential $TC_{gw} = h(ID_{GW} || d_{gw})$ and securely delivers the smartcard SC_i to U_i with the information $\{TID_i, TC_i, h(TC_{gw})\}$. The GW saves $\{TID_i, RID_i\}$ in its database equivalent to registered U_i .

Step 4: Later getting SC_i from the GW, U_i computes $Gen(BIO_i) = (bk_i, pr_i)$, $d'_i = d_i \oplus h(bk_i || ID_i)$, $TC'_i = TC_i \oplus h(d_i || bk_i || ID_i)$, and $RB_i = h(ID_i || bk_i)$. U_i lastly saves the credentials $\{TID_i, RB_i, RID_i, d'_i, TC'_{gw}, h(\cdot), Gen(\cdot), Rep(\cdot), pr_i\}$ SC_i memory, and discards TC_i & $h(TC_{gw})$.

C. Key Management Phase

The establishment of a secret pairwise key among a smart device SD_j GW needs this phase. The pairwise key utilized for two various things: 1) To establish a secure communication between SD_j GW such that information is stored safely at GW, also, establishment of a session key among SD_j and 2) user U_i performed using pairwise key at the login phase. The Major step includes here for gateway calculation with device key pair and session key establishment.

D. User Login and Authentication Phase

Once each session, an authorised user U_i executes this step. After mutual authentication is complete, this process is finished, and the session key is established. To finish this phase successfully, you'll need to take the following steps:

Step 1: $U_i \rightarrow GW: (TID'_i, X_i, Y_i, Z_i, T_1)$ U_i first inserts the issued smart card SC_i into a specific terminal, then enters the

identification ID_i and imprints the personal biometric BIO'_i at the terminal sensor. SC_i then calculates $RB'_i = h(ID_i || bk_i)$ and checks it using the $RB_i = RB'_i$ Condition SC_i ends the session if it does not hold.

Step 2: $GW \rightarrow SD_j: (X_{gw}, Y_{gw}, Z_{gw}, T_2)$ The GW initial task after receiving Msg1 from U_i is to validate the received timestamp's timeliness T_1 using the condition $|T_1^* - T_1| \leq \Delta T$. Where T_1 represents Msg 1 received time and T indicates the highest transmission delay. If the requirement is met, the GW computes $TID_i = TID'_i \oplus h(h(TC_{gw}) || T_1)$ and retrieves TID_i of U_i .

Step 3: $SD_j \rightarrow U_i: (V_j, W_j, T_3)$ When SD_j receives Msg2 at time T_2 , it first verifies the correctness of T_2 in Msg2 using the condition $|T_2^* - T_2| \leq \Delta T$. The phase is terminated if the condition fails. However, SD_j uses its pairwise key $K_{sdjgw} = (K_{gw, sdj})$ shared with the GW to calculate $Z_{sdj} = h(X_{gw} || Y_{gw} || K_{sdjgw} || T_2)$, and then tests the verifying condition $Z_{sdj} = Z_{gw}$. SD_j Completes the phase if the validation is unsuccessful.

Step 4: SC_i validates T_3 in this message by $|T_3^* - T_3| \leq \Delta T$ Once Msg3 is received by U_i at time T_3 . If the condition fails, U_i immediately ends the phase. However, SC_i computes $A_i = V_i \oplus h(h(RID_i || T_1) || T_3) = h(d_{SD_j} || TC_{SD_j} || RID_{SD_j} || K_{sdjgw} || r_{sd} || T_3)$. Finally SC_i accepts the session key SK_{ij} for secret communication in future.

E. Biometric Update Phase

It's a fact that one individual user's biometric information always remains as it is in the beginning, also the changing of that information are occasional. Hence, in this proposed approach we can able to change the biometric for user U_i with Bio_i . The replacement would be as fingerprints or something related to biometric stuff. For that, few initial steps are needed to be performed before the steps to be followed without the interaction of GW to complete the task locally at any time.

Step 1: Three processes must be taken place: firstly, U_i feeds the SC_i smart card to the mentioned terminal, secondly, it pass into ID_i , there at the terminal sensor, printing of BIO_i^{old} biometric should be happened, here, SC_i recovers to the BIO'_i as $bk_i^{old} = Rep(BIO_i^{old}, pr_i)$ from the real biometric provided with $HamDist(BIO_i^{old}, Bio_i)$. SC_i further it calculates $RB_i^{old} = h(ID_i || bk_i^{old})$ and validates with the condition $RB_i^{old} = RB_i$.

Step 2: SC_i , tells U_i to print freshly formed biometric, BIO_i^{new} . Then, SC_i computes as, $Gen(BIO_i^{new}) = (bk_i^{new}, pr_i^{new})$.

Step 3: SC_i computes $d_i = d'_i \oplus h(bk_i^{old} || d_i)$ and $TC_i = TC'_i \oplus h(d_i || bk_i^{old} || ID_i)$, $RID_i = RID'_i \oplus h(bk_i^{old} || d_i)$ and $TC_{gw} = TC'_{gw} \oplus h(RID_i || bk_i^{old} || d_i) = h(TC_{gw})$.

Step 4: SC_i finally replaces the old values RB_i, RID_i, d'_i, TC_i and pr_i with the current values $RB_i^{new}, RID_i^{new}, d_i^{new}, TC_i^{new}, TC_{gw}^{new}$, and pr_i^{new} , respectively, in its memory.

F. Security analysis for RCBP2U-AS

The exploration of several known attacks, the current work is an security analysis of RCBP2U-AS method in detail.

1) *Redeem from verification table or password*: The value of q is kept as a secret by the base station, which the user uses to register. In this system, there is no need to keep a password table. As a result, the RCBP2U-AS method is secure against password alteration attacks.

2) *Replay attack*: In order to get access to the network, the adversary must replay the authentication session. The login process in RCBP2U-AS includes a random nonce and timestamps. Whenever the new entry occurs, this technique checks for the base station and the user. The corresponding adversary should take the random numbers and estimate the value of A_6 , A_8 and with freshly created time stamp as to process the attack. In the case the intruder must find out A_3 first to discover the A_6 . To complete the process (A_3) the attacker should set the value as Z and password which recognized only by the respective base station. As a result of this, it's quite obvious that RCBP2U-AS is considered to be attack free.

3) *Offline password guessing attack*: The password usually coded as $(h(PW_i))$ format in that way it's not going to be easier for hackers to guess as it's a one directional hash property. At the time of login phase, it's communicated at A_2 , A_6 and A_9 also it stored in A_2 format in the smart card. The adversary must know the values of R_{sc} , q , and Z in order to guess the actual password, but it is impossible to predict all three values at the same time.

4) *Password modification attack*: As mentioned in the previous part of the offline password guessing attack, the approach does not keep any of the passwords neither verification figure. As a result, RCBP2U-AS is resistant to modification.

5) *User anonymity*: Once the intruder gets the access of the device and login authentication, he immediately tries to hack the login credential by identifying the communication among the messages. However, hash function and timestamp value protect these communications; random nonce varies per two logins. As a result, an attacker will have a difficult time gaining access to the user's credentials. Also, the proposed scheme ensures user privacy.

IV. RESULT AND DISCUSSION

The current approach is encoded inside HLPSP and also evaluated at the same time. According to the simulation outcomes of AVISPA Tool, RCBP2U-AS considered to be safest. As a result of this, the output is then backup to the automatic analysis.

RCBP2U-AS throughput values observed as, 183.13 bits per second, 286.84 bps, 244.73 bps 244.11 bps. During the simulation various messages exchanged between the entities were observed and compared with existing techniques. Moreover, it utilizes very tiny messages, also produces the results with minimum hops of communication for the login phase compared to other related techniques. The end-to-end delay presented as $\sum_{i=1}^{N_p} (T_{rec_i} - T_{send_i}) N_p$. here N_p is the

TABLE II
MESSAGE EXCHANGES BETWEEN ENTITIES UTILIZED SIMULATION

Messages exchanged between entities	Challa et al. [5]	Turkanovic et al. [15]	Amin et al. [3]	Proposed Method
$U_i \rightarrow GW/S_{i,m}$	992 bits	672 bits	672 bits	672 bits
$GW \rightarrow SD_j/Cs$	1024 bits	1024 bits	1184 bits	512 bits
$SD_j \rightarrow GW/S_{i,m}$	-	576 bits	640 bits	-
$GW \rightarrow U_i$	-	448 bits	320 bits	-
$SD_j \rightarrow U_i$	512 bits	-	-	352 bits

total packets, T_{send_i} and T_{rec_i} sent and receive the packets with i . The throughput packet loss ratio encountered to be as 0.017, 0.030, 0.041 0.033. However, the rate is higher for RCBP2U-AS than others, with better functionality and enhanced security. These parameters ensure the efficiency of the system with robust mechanism.

V. CONCLUSIONS

Recently, emerging solutions for authentication scheme using multifactor have gained extensive attention. Particularly it's easy to generate session key with device compatibility in an IoT environment, but at the same time it's difficult to break the accessibility of any remote device in listening the content. So, we proposed Rabin cryptosystem based biometric specifically on privacy preserving factors. Moreover, the simulation outcome shows that the scheme could manage all major assaults as in replay, man-in-the-middle attacks and other active passive attacks. To comprehensively analyze the impact of attacks on the system, we experimented the test cases using AVISPA tool. The recommended scheme enables mutual authentication session key agreement features based on BAN logic. The technique moreover endures on multiple dangerous acts, as per the analysis of privacy validation. In particular to compete the suitable existing techniques, the current method has been evaluated and tested against execution time, memory cost, end-to-end delay, packet loss and a lot more. The system is long sustainable to work in real time data platforms for the futuristic works in the domain of agriculture sector, military services, healthcare etc. Also, it has the ability to contain and execute massive information. Moreover, RCBP2U-AS can be utilized to enrich the system functionality to mitigate the risks and diminish the communication and computational prices.

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